Linear Algebra 2025 Fall Midterm

2025.11.14 13:30–16:00 total 100 points

General information. Cheatsheets and calculators are not allowed.

Side note. All *Side notes* did not appear on the exam paper, but the **Hints** did. Moreover, some problems may not be described very clearly, but they are similar to the descriptions in the exam. Problem 6 was a numerical calculation problem and the numbers cannot be recalled, but they are actually trivial and can be filled in arbitrarily.

Problem 01

Consider a system of linear equations:

$$\begin{aligned}(a_1+b)x_1+a_2x_2+\cdots+a_nx_n&=0,\\ a_1x_1+(a_2+b)x_2+\cdots+a_nx_n&=0,\\ &\vdots\\ a_1x_1+a_2x_2+\cdots+(a_n+b)x_n&=0,\end{aligned}$$

where $\sum_{i=1}^{n} a_1 \neq 0$.

(1) Find the conditions for $a_1,\ldots,a_n,b$ such that the system has only the zero solution.

(2) Find the conditions for $a_1,\ldots,a_n,b$ such that the system has non-zero solutions. In this case, find the solution sets.

Problem 02

Let $T: V \to W$ be a linear transformation. Prove that $\operatorname{nullity} T + \operatorname{rank} T = \dim V$.

Problem 03

Suppose V is an n-dimensional vector space. Let $T: V \to V$ be a linear transformation, and W an r-dimensional subspace of V that is invariant under T. Prove that there exists a basis β of V such that $[T]_\beta$ takes the form

$$\begin{pmatrix} A & B \\ O & C \end{pmatrix},$$

where A is an $r \times r$ square matrix, and O is the $(n-r)\times r$ zero matrix.
Hint: W is an invariant subspace means that $T(W) \subseteq W$.

Problem 04

(1) Define the Cartesian product of vector spaces:

$$V_1 \times \cdots \times V_m := \{(v_1, \ldots, v_m) : v_1 \in V_1, \ldots, v_m \in V_m\}.$$

It can be verified that $V_1 \times \cdots \times V_m$ is a vector space (*Side note: verification was given in the exam, but omitted here for simplicity*). Find the dimension of $V_1 \times \cdots \times V_m$.

(2) Prove that $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ is isomorphic to $\mathcal{L}(V_1 \times \cdots \times V_m, W)$.

Problem 05

Suppose A, B, and $A + B$ are all $n \times n$ invertible matrices. Prove that $A^{-1} + B^{-1}$ is also invertible and find its inverse matrix.

Problem 06

Side note: numerical calculation problem, but the numbers are irretrievable.

(1) Suppose $T : \mathbb{R}^4 \to \mathbb{R}^4$ is a linear transformation with

$$T(e_1) = \cdots, \quad T(e_2) = \cdots, \quad T(e_3) = \cdots, \quad T(e_4) = \cdots$$

Side note: you can fill in four linearly independent vectors by yourself (there are about six zeroes in the four vectors). Calculate the rank of T and determine if it is invertible.

(2) Find the dimension and a basis of $R(T)$ and $N(T)$, respectively.

(3) Suppose there is a new basis β:

$$\{\cdots, \cdots, \cdots, \cdots\}.$$

Side note: again, any four linearly independent vectors are okay (there are about eight zeroes in the four vectors). Calculate $[T]_\beta$.

Problem 07

Let $T : V \to W$ be a linear transformation and $T^t : W^* \to V^*$ its transpose. Suppose there exists $\phi \in W^*$ such that $N(T^t) = \text{span}(\phi)$. Prove that $R(T) = N(\phi)$.
Hint: the definition of T^t was given in class.

Problem 08

Suppose A is an $n \times n$ matrix with rank r $(1 \leq r \leq \frac{n}{2})$ and $A^2 = 0$. Prove that there exists an invertible matrix P such that

$$A = P \begin{pmatrix} O & I_r & O \\ O & O & O \end{pmatrix} P^{-1},$$

where I_r is the $r \times r$ identity matrix, and O are zero matrices.